

Surname	Centre Number	Candidate Number
Other Names		0

**GCSE****3430UB0-1**

S19-3430UB0-1

WEDNESDAY, 12 JUNE 2019 – MORNING**SCIENCE (Double Award)****Unit 2: CHEMISTRY 1
HIGHER TIER****1 hour 15 minutes****ADDITIONAL MATERIALS**

In addition to this examination paper you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen. Do not use correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question **6** is a quality of extended response (QER) question where your writing skills will be assessed.

The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.

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Question	Maximum Mark	Mark Awarded
1.	8	
2.	7	
3.	8	
4.	9	
5.	5	
6.	6	
7.	9	
8.	8	
Total	60	

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Answer all questions.

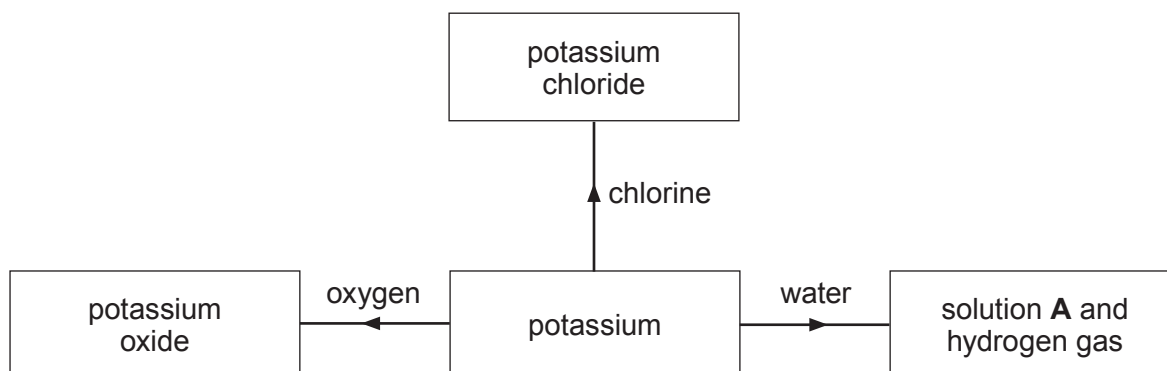
1. (a) The table gives information about some elements.

Element	Electronic structure	Group	Period
oxygen	2,6	6	2
chlorine		7	3
.....	2,8,5	5	3
potassium	2,8,8,1	1	

Complete the table.

[3]

- (b) The flow chart shows some of the reactions of potassium.



- (i) State **one** observation you would make when potassium reacts with water. [1]

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- (ii) Apart from wearing gloves and safety goggles, give **one** safety precaution that should be taken when adding potassium to water. [1]

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(iii) Give the formula of solution **A**.

[1]

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(iv) Suggest a value for the pH of solution **A**.

[1]

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(v) Name a Group 1 metal that is **more** reactive than potassium.

[1]

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2. (a) The Earth's early atmosphere around 4 000 million years ago contained mainly carbon dioxide and water vapour produced by volcanoes.

(i) Explain why the large percentage of water vapour in the Earth's atmosphere decreased over geological time. [2]

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(ii) Give **two** reasons why the percentage of carbon dioxide in the Earth's atmosphere has decreased over geological time. [2]

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(b) During the last 250 years the percentage of carbon dioxide in the Earth's atmosphere has increased from 0.03 % to 0.04 %. This has led to increased global warming. Give **one** reason for this increase and explain why global warming is a cause for concern. [2]

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(c) Ammonia present in the Earth's early atmosphere reacted with oxygen to produce nitrogen and water vapour. Complete the balancing of the symbol equation for this reaction. [1]



3. Some Year 10 students were given three unknown white solids – **A**, **B** and **C**. They carried out a series of flame tests and silver nitrate tests to identify the solids. Their results are shown in the table.

Solid	Observations	
	Flame test	Silver nitrate test
A	apple-green flame	cream precipitate
B	red flame	white precipitate
C	yellow flame	yellow precipitate

- (a) Name solids **A**, **B** and **C**. [3]

A

B

C

- (b) Complete and balance the symbol equation for the reaction between magnesium chloride and silver nitrate. [2]



- (c) 0.103 g of silver nitrate, AgNO_3 , was used to make up a solution.

Calculate the number of moles of silver nitrate in this mass. Give your answer in **standard form**. [3]

$$A_r(\text{Ag}) = 108$$

$$A_r(\text{N}) = 14$$

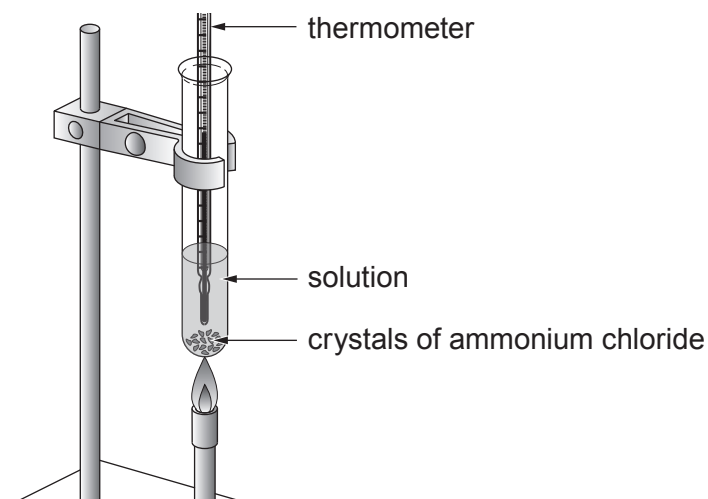
$$A_r(\text{O}) = 16$$

Number of moles = mol



4. A student investigates the solubility of ammonium chloride by adding different masses to 10 g of water.

He uses the apparatus shown.



10 g of water is placed in a boiling tube and 3.0 g of ammonium chloride is added.

The tube is heated until all the solid dissolves.

The tube is allowed to cool.

The temperature at which solid ammonium chloride first appears is recorded.

The experiment is repeated using different masses of ammonium chloride.

The results are shown in the table.

Mass of ammonium chloride in 10 g of water (g)	3.0	3.3	4.1	5.2	5.9	6.6
Temperature at which solid ammonium chloride first appears (°C)	4	10	30	52	68	80

- (a) What practical problem is the student likely to come across in finding the first two results? Suggest how this problem might be overcome. [2]

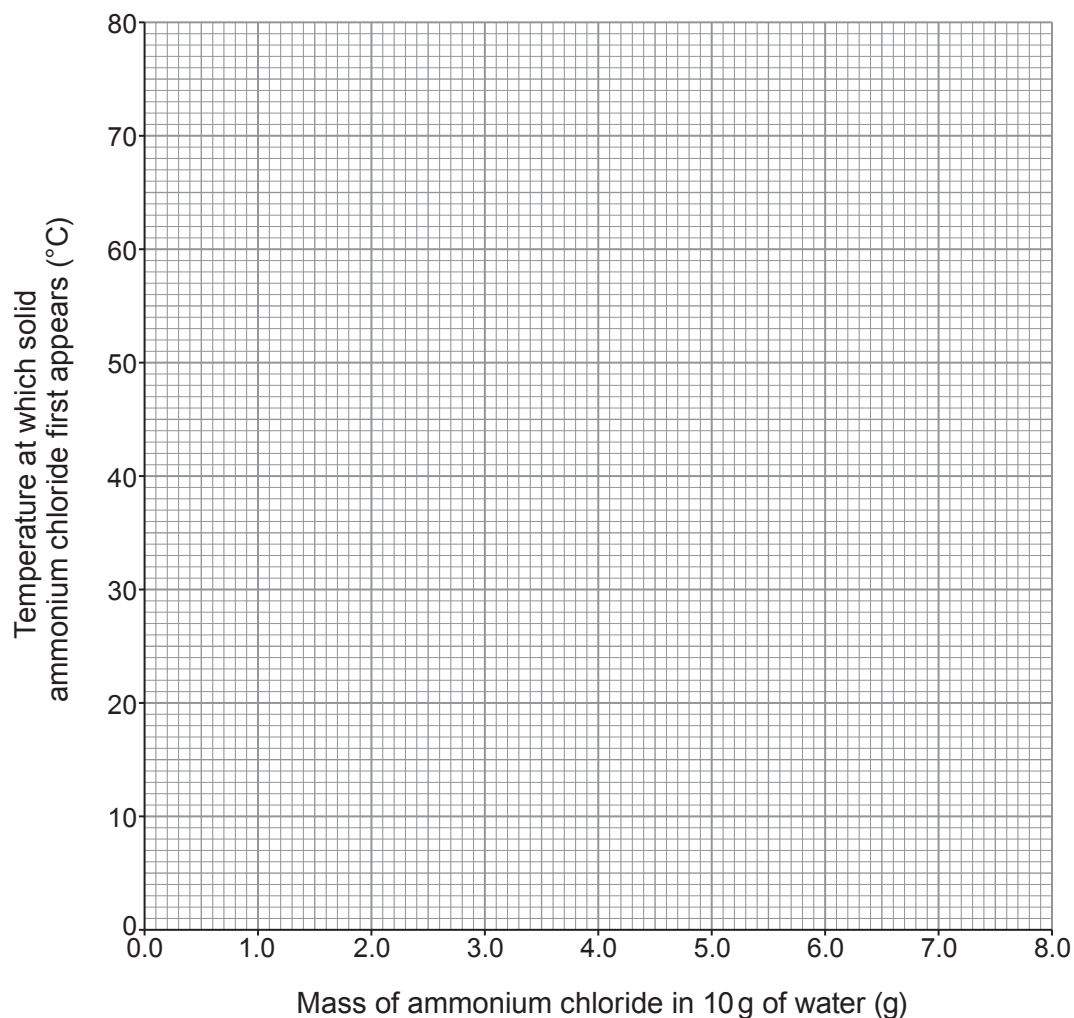
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- (b) (i) On the grid below, plot the temperature at which solid ammonium chloride first appears against the mass of ammonium chloride in 10 g of water. Draw a suitable line. [3]



- (ii) The student is given a boiling tube containing 5.0 g of ammonium chloride in 10 g of water. He stirs the ammonium chloride in the water and heats it to a temperature of 45°C.

State whether all the ammonium chloride dissolves. Give a reason for your answer. [1]

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- (c) The student is asked to use a **different** method to find the exact solubility of another compound in water at room temperature. He knows that it has a value of approximately 7g per 100g of water at this temperature.

He is given a 5.0g sample of the compound and common laboratory equipment but **no heating apparatus**.

Describe how he would carry out his method and how he would find the solubility. [3]

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5. (a) A bottle contains a mixture of liquids **E** and **F**. Liquid **E** has a boiling point of 57 °C and liquid **F** has a boiling point of 95 °C.

Describe the process of distillation and explain why it can be used to separate these liquids. [3]

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- (b) One molecule of liquid **E** contains **two** oxygen atoms. The percentage by mass of oxygen in liquid **E** is 43.2 %.

Use the following equation to calculate the relative molecular mass (M_r) of liquid **E**. [2]

$$\frac{\text{mass of oxygen}}{M_r} \times 100 = 43.2$$

$$A_r(\text{O}) = 16$$

$$M_r = \dots\dots\dots$$

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6. Explain how the processes of boiling and adding washing soda remove hardness from water. Include equations in your answer. [6 QER]

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7. (a) (i) Chlorine is a non-metal found in Group 7 of the Periodic Table. When it is bubbled into a solution of potassium iodide there is a colour change from pale green to brown. Explain why this reaction occurs. [2]

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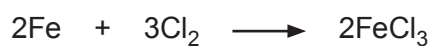
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- (ii) Write the balanced symbol equation for the reaction between chlorine and potassium iodide. [2]

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- (b) The symbol equation for the reaction between iron and chlorine is as follows.



Calculate the mass of chlorine needed to react with 1.32 g of iron. [3]

$$A_r(\text{Fe}) = 56 \qquad A_r(\text{Cl}) = 35.5$$

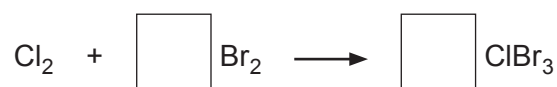
Mass of chlorine = g



- (c) (i) Under certain conditions, Group 7 elements will react with each other to produce new compounds.

When chlorine is reacted with bromine, chlorine tribromide is made.

Balance the symbol equation for this reaction. [1]



- (ii) A chemist calculated that if she reacted 7.00 g of chlorine with an excess of bromine, the theoretical mass of chlorine tribromide produced is 27.55 g.

However, when she carried out the experiment using 7.00 g of chlorine the mass of chlorine tribromide obtained was 21.34 g.

Calculate the percentage yield of chlorine tribromide. [1]

Percentage yield = %



8. Hydrogen peroxide solution, H_2O_2 , is used in commercial stain removers.

A GCSE class investigated how effective four stain removers are at removing stains. Stain removers **A**, **B**, **C** and **D** contain different concentrations of hydrogen peroxide.

The students tested how effective each one is at removing an identical oil stain from four towels.

Their findings are outlined below.

Stain remover A

Working temperature	= 50 °C
Cost per 100 cm ³	= 99p
Time to remove stain	= 40 min
Volume needed	= 20 cm ³

Stain remover B

Working temperature	= 30 °C
Cost per 100 cm ³	= £1.99
Time to remove stain	= 40 min
Volume needed	= 10 cm ³

Stain remover C

Working temperature	= 20 °C
Cost per 100 cm ³	= £2.49
Time to remove stain	= 20 min
Volume needed	= 5 cm ³

Stain remover D

Working temperature	= 30 °C
Cost per 100 cm ³	= £1.49
Time to remove stain	= 30 min
Volume needed	= 10 cm ³

- (a) In carrying out this investigation, which variables were kept the same in order to get valid results? Tick (✓) the correct answer. [1]

type of oil used, towel material and volume of hydrogen peroxide

☐

type of oil used, towel material and temperature of stain remover

☐

type of oil used and towel material

☐

type of oil used, towel material and cost of stain remover

☐


- (b) Tick (✓) **all** of the statements which **could** explain why stain remover **A** has to be heated to 50 °C before it removes the stain. [1]

it is the cheapest stain remover

☐

it is heat resistant

☐

it has a low concentration of hydrogen peroxide

☐

it takes a long time to work

☐

- (c) The students found that stain removers **B** and **D** used the same volume and worked best at the same temperature.

Assuming that they have the same hydrogen peroxide concentration, suggest a possible reason why **D** removes the stain more quickly than **B**. [1]

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(d) One student went on to investigate the decomposition of hydrogen peroxide.

The equation for the reaction is as follows.

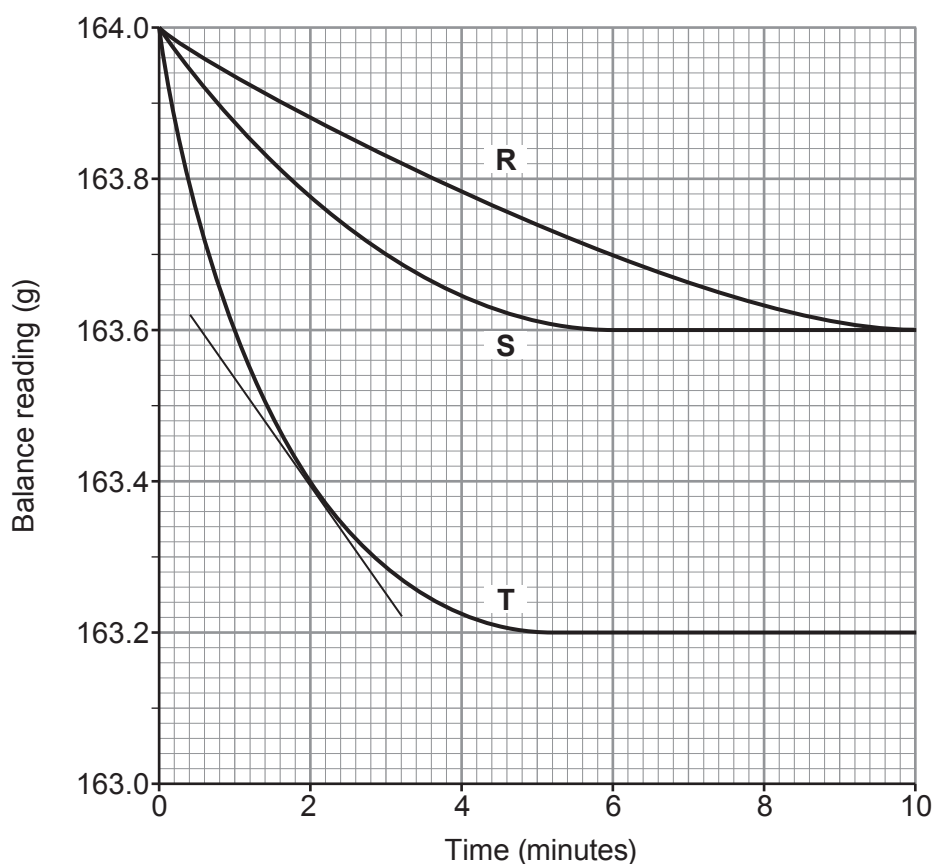


The student investigated the effect of changing the concentration of the hydrogen peroxide solution on the rate of the reaction. She used manganese dioxide as a catalyst in each experiment.

This is the method she used.

- Pour 50 cm³ of hydrogen peroxide solution of concentration **R** into a conical flask on a digital balance.
- Add 1 g of catalyst and place some cotton wool loosely in the neck of the flask. Record the balance reading and immediately start a stopwatch.
- Record the balance reading every minute until the mass no longer changes.
- Carry out the experiment twice more using hydrogen peroxide of different concentrations, **S** and **T**.

Her results are plotted on the grid below.



- (i) Using the tangent shown on the graph, calculate the rate of reaction for concentration **T** at 2 minutes. Show your working. [2]

Rate at 2 minutes = g/minute

- (ii) The initial rate for concentration **S** is half the initial rate for concentration **T**. Explain this difference in rate in terms of the particle theory. [3]

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END OF PAPER



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FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^-
ammonium	NH_4^+	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^-
calcium	Ca^{2+}	fluoride	F^-
copper(II)	Cu^{2+}	hydroxide	OH^-
hydrogen	H^+	iodide	I^-
iron(II)	Fe^{2+}	nitrate	NO_3^-
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^+	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		





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THE PERIODIC TABLE

1 2

Group

3 4 5 6 7 0

1	H	1
	Hydrogen	

7	Li	9	Be
	Lithium		Beryllium
	3		4
23	Na	24	Mg
	Sodium		Magnesium
	11		12
39	K	40	Ca
	Potassium		Calcium
	19		20
86	Rb	88	Sr
	Rubidium		Strontium
	37		38
133	Cs	137	Ba
	Caesium		Barium
	55		56
223	Fr	226	Ra
	Francium		Radium
	87		88
		227	Ac
			Actinium
			89

11	B	12	C	14	N	16	O	19	F	20	Ne
	Boron		Carbon		Nitrogen		Oxygen		Fluorine		Neon
	5		6		7		8		9		10
27	Al	28	Si	31	P	32	S	35.5	Cl	40	Ar
	Aluminium		Silicon		Phosphorus		Sulfur		Chlorine		Argon
	13		14		15		16		17		18
70	Ga	73	Ge	75	As	79	Se	80	Br	84	Kr
	Gallium		Germanium		Arsenic		Selenium		Bromine		Krypton
	31		32		33		34		35		36
115	In	119	Sn	122	Sb	128	Te	127	I	131	Xe
	Indium		Tin		Antimony		Tellurium		Iodine		Xenon
	49		50		51		52		53		54
204	Tl	207	Pb	209	Bi	210	Po	210	At	222	Rn
	Thallium		Lead		Bismuth		Polonium		Astatine		Radon
	81		82		83		84		85		86

65	Zn	63.5	Cu	59	Ni	59	Co	56	Fe	55	Mn
	Zinc		Copper		Nickel		Cobalt		Iron		Manganese
	30		29		28		27		26		25
112	Cd	108	Ag	106	Pd	103	Rh	101	Ru	99	Tc
	Cadmium		Silver		Palladium		Rhodium		Ruthenium		Technetium
	48		47		46		45		44		43
201	Hg	197	Au	195	Pt	192	Ir	190	Os	186	Re
	Mercury		Gold		Platinum		Iridium		Osmium		Rhenium
	80		79		78		77		76		75
184	W	181	Ta	181	Ta	184	W	179	Hf	179	Hf
	Tungsten		Tantalum		Tantalum		Tungsten		Hafnium		Hafnium
	74		73		73		74		72		72
139	La	137	Ba	139	La	184	W	179	Hf	179	Hf
	Lanthanum		Barium		Lanthanum		Tungsten		Hafnium		Hafnium
	57		56		57		74		72		72
227	Ac	226	Ra	227	Ac	186	Re	190	Os	186	Re
	Actinium		Radium		Actinium		Rhenium		Osmium		Rhenium
	89		88		89		75		76		75

Key

Ar	relative atomic mass
Symbol	
Name	
Z	atomic number